KAIZEN MATHS

Lesson starter

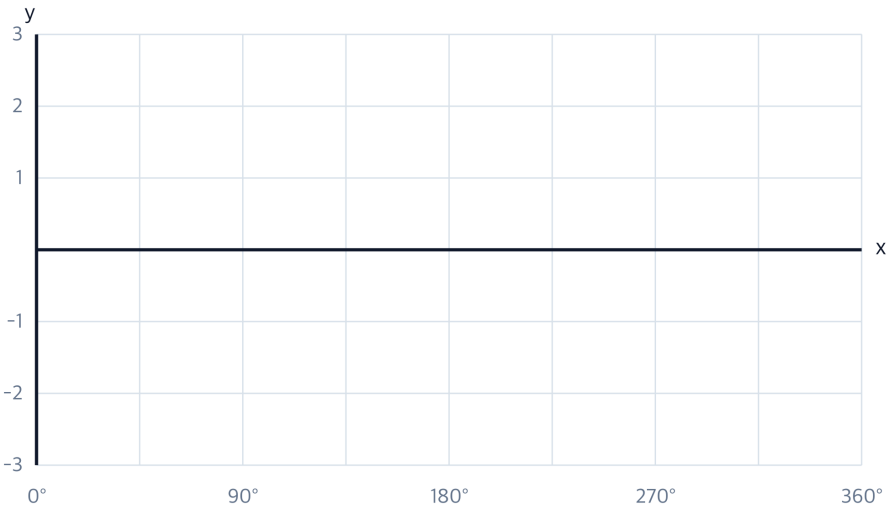
Trigonometric Graphs and Transformations

Combined Transformations

Name _____

SA1

Sketch each graph and state the phase shift.



Sketch y = cos(x - 90°) and state the phase shift.

Show your working

KAIZEN MATHS

Lesson starter

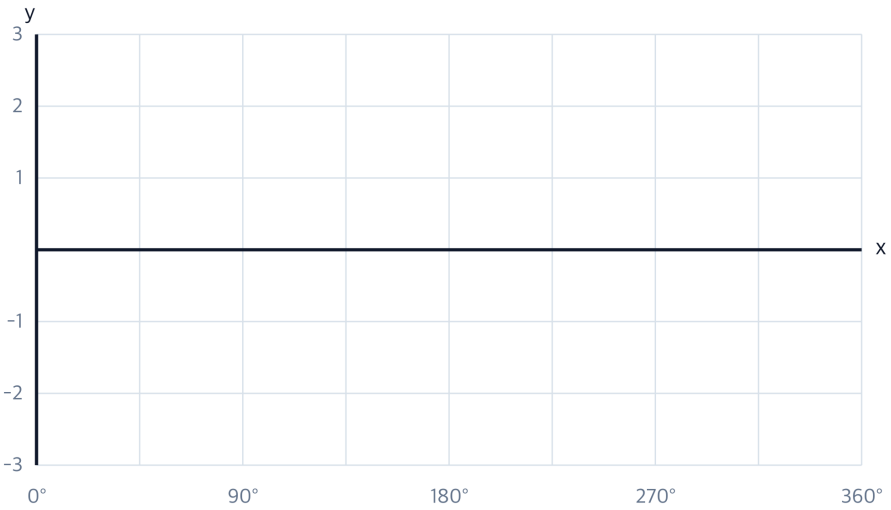
Trigonometric Graphs and Transformations

Combined Transformations

Name _____

SB1

Use amplitude, period, phase shift, and midline to sketch each graph.



Sketch y = 3cos(2(x)) for 0° ≤ x ≤ 360°.

Show your working

KAIZEN MATHS

Lesson starter

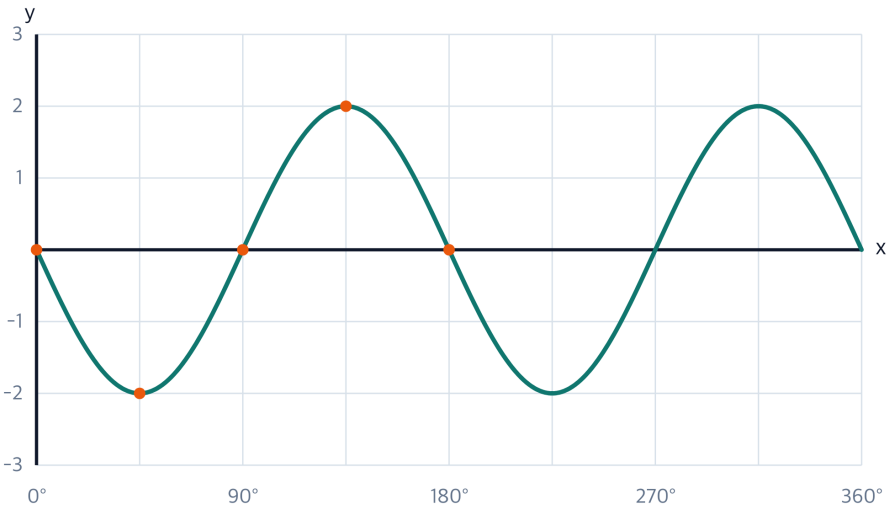
Trigonometric Graphs and Transformations

Combined Transformations

Name _____

SC1

Describe the transformations represented by each equation.



Describe the graph transformations represented by y = -2sin(2(x)).

ALG-020E-L3 · SC

Show your working

KAIZEN MATHS

Lesson starter

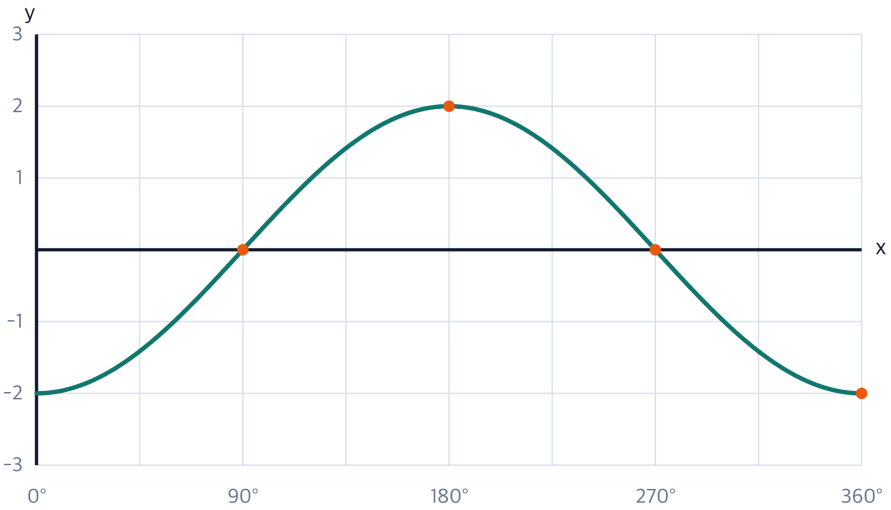
Trigonometric Graphs and Transformations

Combined Transformations

Name _____

SD1

Use the graph features to identify a possible equation.



Find a possible equation for the graph shown.

ALG-020E-L3 · SD

Show your working

KAIZEN MATHS

Exit ticket

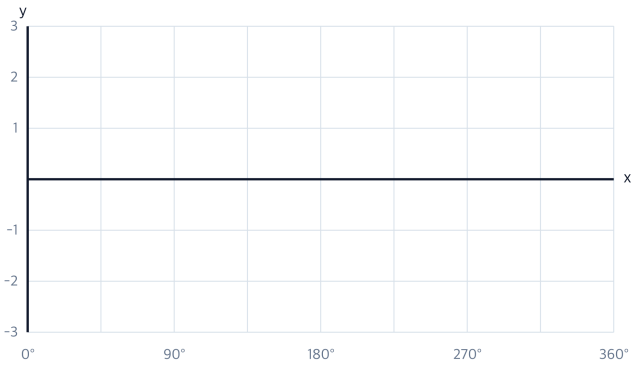
Trigonometric Graphs and Transformations

Combined Transformations

Name

EA1

Sketch each graph and state the phase shift.



Sketch y = sin(x + 90°) and state the phase shift.

Show your working

KAIZEN MATHS

Exit ticket

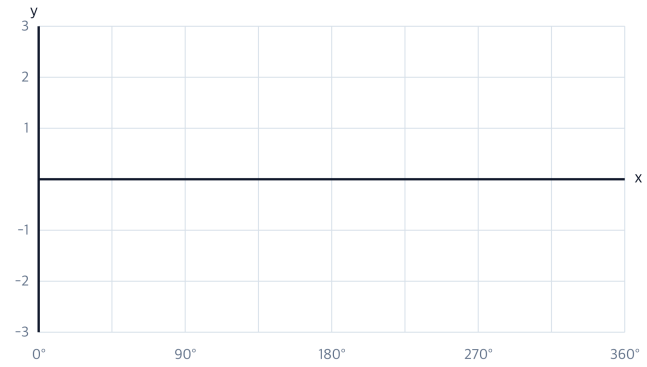
Trigonometric Graphs and Transformations

Combined Transformations

Name

EB1

Use amplitude, period, phase shift, and midline to sketch each graph.



Sketch y = 2sin(2(x + 90°)) - 1 for 0° ≤ x ≤ 360°.

Show your working

KAIZEN MATHS

Exit ticket

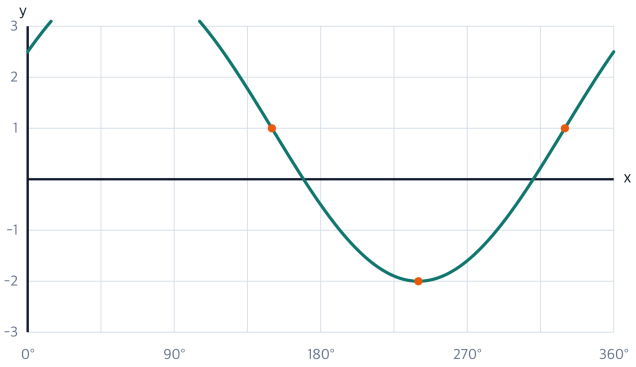
Trigonometric Graphs and Transformations

Combined Transformations

Name

EC1

Describe the transformations represented by each equation.



Describe the graph transformations represented by y = 3cos(x - 60°) + 1.

Show your working

KAIZEN MATHS

Exit ticket

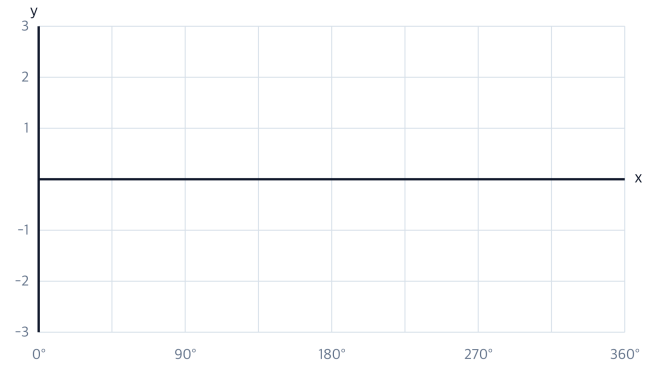
Trigonometric Graphs and Transformations

Combined Transformations

Name

EA1

Sketch each graph and state the phase shift.



Sketch y = sin(x + 90°) and state the phase shift.

Show your working

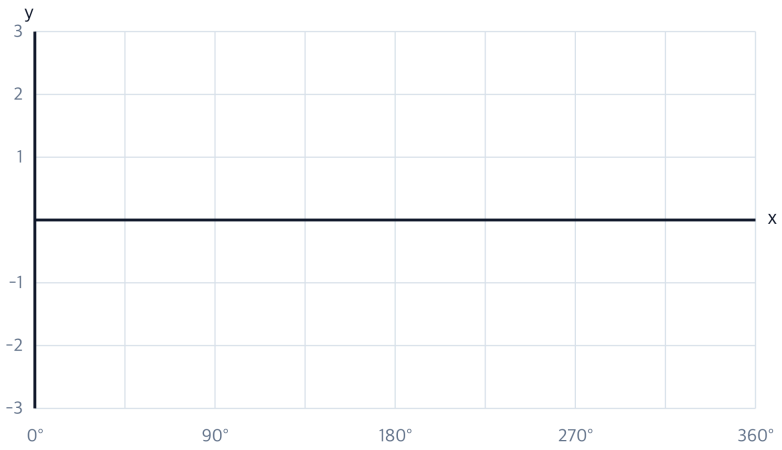
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

ALG-020E-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Sketch each graph and state the phase shift.



Sketch y = cos(x - 90°) and state the phase shift.

Answer:

Phase shift: 90° right.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 90° right.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	1
Period	360°
Midline	$y = 0$
Phase shift	90° right

Use the transformed key points to guide the sketch.

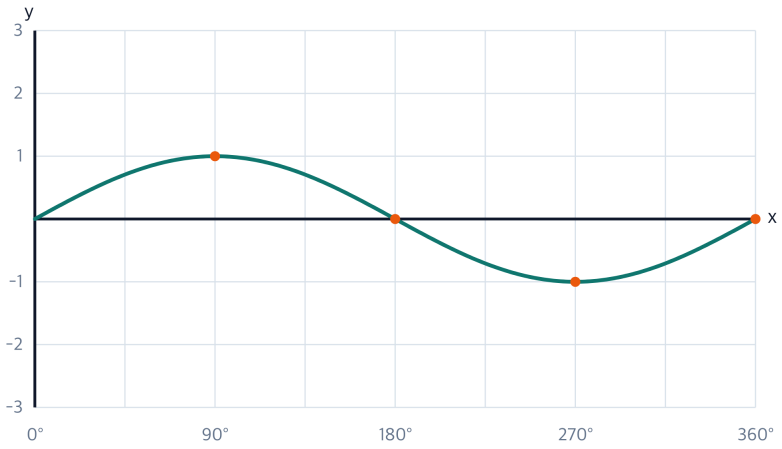
x	y
90°	1
180°	0
270°	-1
360°	0

Teacher answers & working

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SA1 (continued)



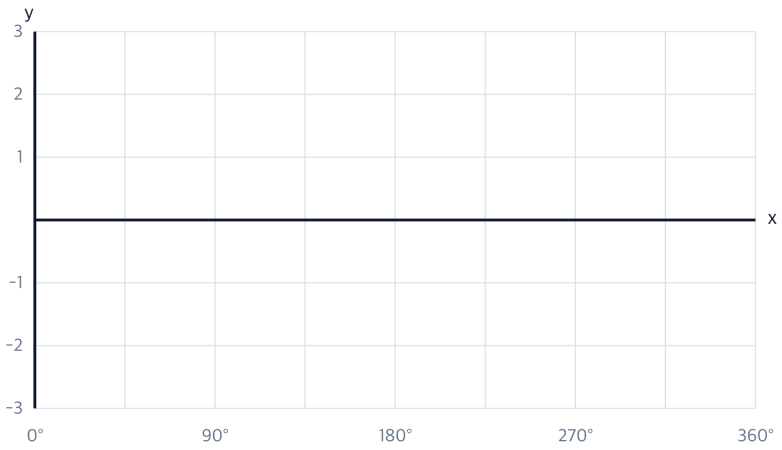
Teacher answers & working

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SB1

Use amplitude, period, phase shift, and midline to sketch each graph.



Sketch y = 3cos(2(x)) for 0° ≤ x ≤ 360°.

Answer:

Amplitude 3, period 180°, midline y = 0, phase shift none.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 3, so the amplitude is |3| = 3.

The coefficient inside the bracket is 2, so the period is 360° ÷ 2 = 180°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	3
Period	180°
Midline	y = 0
Phase shift	none

Use the transformed key points to guide the sketch.

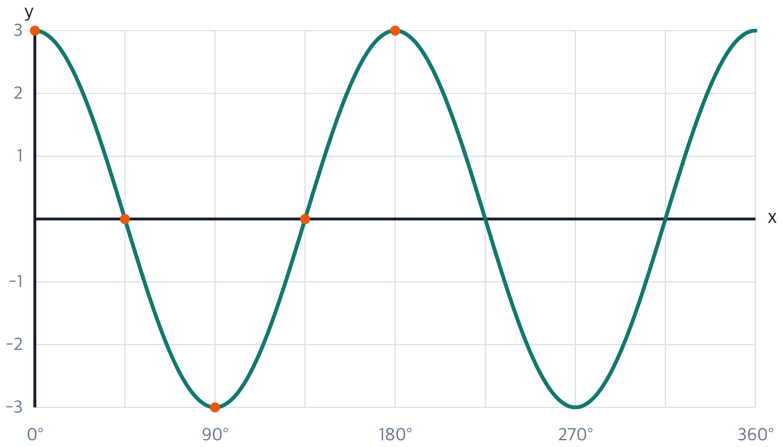
x	y
0°	3
45°	0
90°	-3
135°	0
180°	3

Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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SB1 (continued)



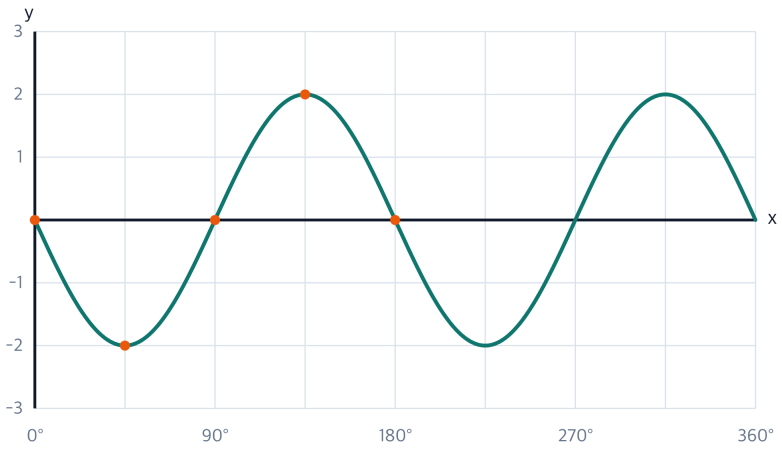
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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SC1

Describe the transformations represented by each equation.



Describe the graph transformations represented by y = -2sin(2(x)).

Answer:

Vertical scale factor -2; period 180°; phase shift none; midline y = 0.

Working:

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is -2, so the amplitude is |-2| = 2.

The coefficient inside the bracket is 2, so the period is 360° ÷ 2 = 180°.

The graph is shifted 0° horizontally.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	2
Period	180°
Midline	y = 0
Phase shift	none

Use the transformed key points to guide the sketch.

x	y
0°	0
45°	-2
90°	0
135°	2
180°	0

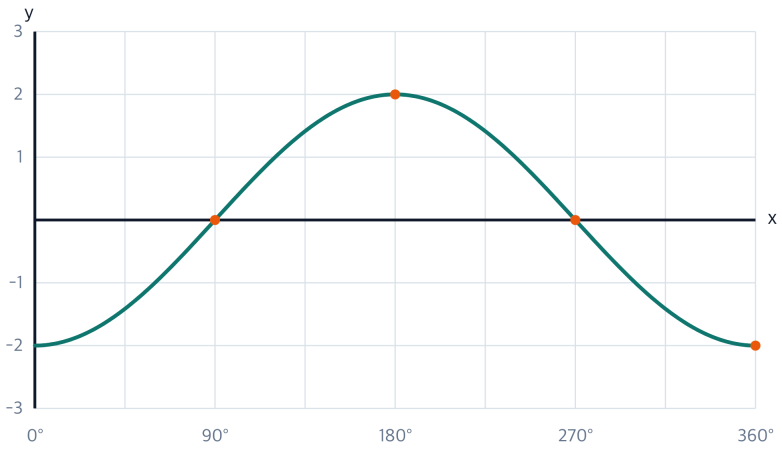
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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SD1

Use the graph features to identify a possible equation.



Find a possible equation for the graph shown.

Answer:

$y = 2\sin(x - 90°)$

Working:

Read the midline from the graph: y = 0.

Read the amplitude from the distance between the midline and a maximum or minimum: amplitude = 2.

Read the period from one complete cycle: period = 360°.

Use sin because the graph follows the sin starting pattern after the phase shift.

A possible equation is y = 2sin(x - 90°).

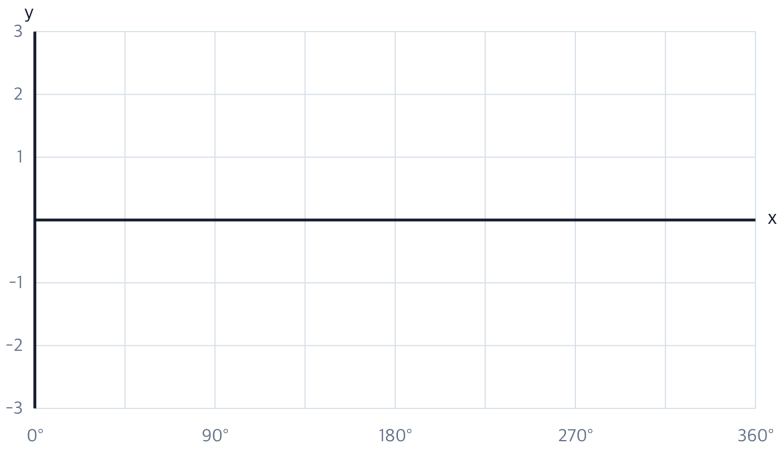
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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EA1

Sketch each graph and state the phase shift.



Sketch y = sin(x + 90°) and state the phase shift.

Answer:

Phase shift: 90° left.

Working:

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is 1, so the amplitude is |1| = 1.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 90° left.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

Amplitude	1
Period	360°
Midline	$y = 0$
Phase shift	90° left

Use the transformed key points to guide the sketch.

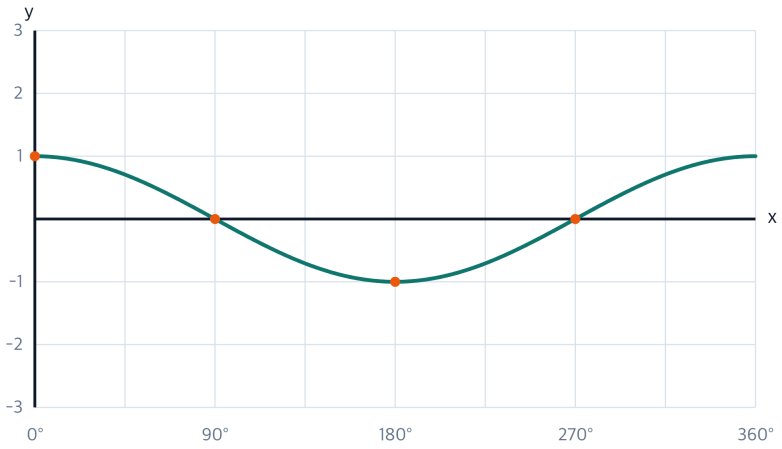
x	y
0°	1
90°	0
180°	-1
270°	0

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EA1 (continued)



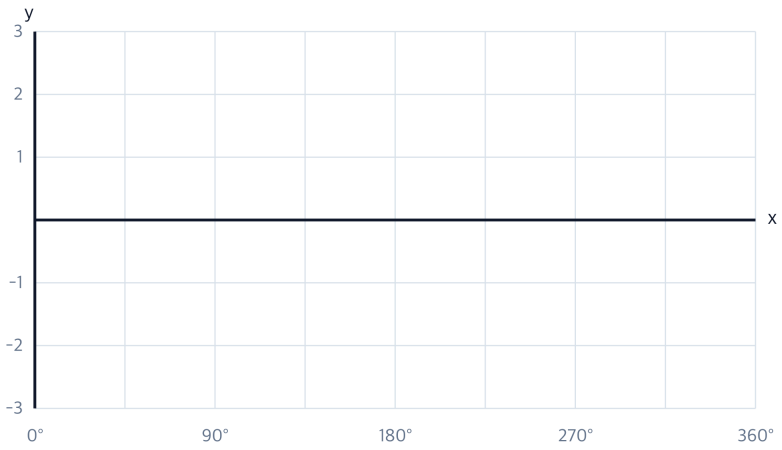
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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EB1

Use amplitude, period, phase shift, and midline to sketch each graph.



Sketch y = 2sin(2(x + 90°)) - 1 for 0° ≤ x ≤ 360°.

Answer:

Amplitude 2, period 180°, midline y = -1, phase shift 90° left.

Working:

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is 2, so the amplitude is |2| = 2.

The coefficient inside the bracket is 2, so the period is 360° ÷ 2 = 180°.

The graph is shifted 90° left.

The - 1 outside the function gives the midline y = -1.

Summarise the graph features.

Amplitude	2
Period	180°
Midline	y = -1
Phase shift	90° left

Use the transformed key points to guide the sketch.

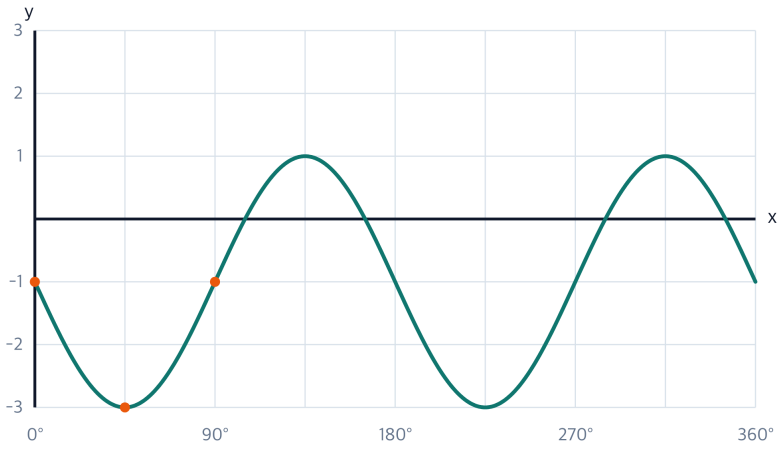
x	y
0°	-1
45°	-3
90°	-1

Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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EB1 (continued)



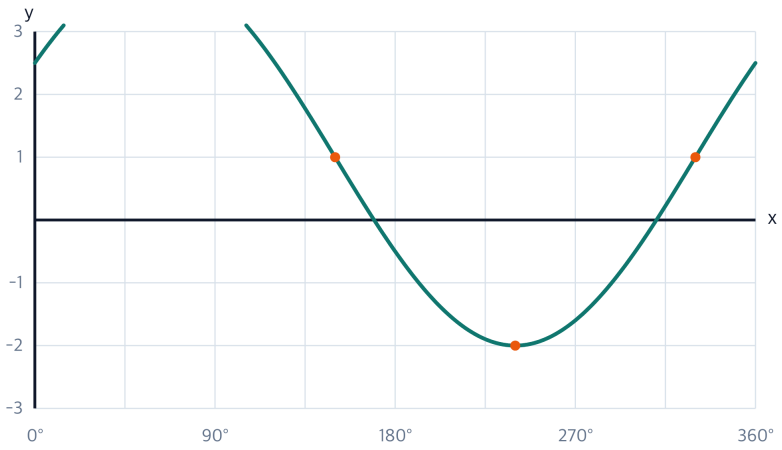
Teacher answers & working

Trigonometric Graphs and Transformations · Combined Transformations

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EC1

Describe the transformations represented by each equation.



Describe the graph transformations represented by y = 3cos(x - 60°) + 1.

Answer:

Vertical scale factor 3; period 360°; phase shift 60° right; midline y = 1.

Working:

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is 3, so the amplitude is |3| = 3.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 60° right.

The + 1 outside the function gives the midline y = 1.

Summarise the graph features.

Amplitude	3
Period	360°
Midline	y = 1
Phase shift	60° right

Use the transformed key points to guide the sketch.

x	y
60°	4
150°	1
240°	-2
330°	1